Recovery From High-Intensity Training Sessions in Female Soccer Players

Jesper Sjökvist,[1,2] Matthew C. Laurent,[3] Mark Richardson,[4] Matt Curtner-Smith,[4]
Hans-Christer Holmberg,[1,2] and Philip A. Bishop[4]

[1]Department of Health Sciences, Swedish Winter Sports Research Center, Mid Sweden, University, Östersund, Sweden; [2]Swedish Olympic Committee, Stockholm, Sweden; [3]Department of Kinesiology, St. Ambrose University, Davenport, Iowa; and [4]Department of Kinesiology, University of Alabama, Tuscaloosa, Alabama

Abstract

Sjökvist, J, Laurent, CM, Richardson, M, Curtner-Smith, M, Holmberg, H-C, and Bishop, PA. Recovery from high-intensity training sessions in female soccer players. *J Strength Cond Res* 25(6): 1726–1735, 2011—This study quantified the performance recovery time requirements after training sessions using high-intensity soccer drills with and without the ball in National Collegiate Athletic Association Division I female soccer players. Recovery time periods (24, 48, 72 hours of rest) from high-intensity soccer training sessions using drills with and without the ball were evaluated. Markers of recovery were each individual's performance relative to baseline performance in countermovement jump (CMJ) height, 5 bound jumps for distance (5BT), 20-m sprint (20SP), session rating of perceived effort (S-RPE), and heart rate (HR). Repeated-measures analysis of variance revealed a significant difference in CMJ performance ($p < 0.04$) and S-RPE ($p < 0.02$) after 24 hours of rest but not at 48 or 72 hours compared to baseline. There were no significant differences in 20SP, 5BT, or HR after 24, 48, or 72-hour recovery ($p > 0.05$). Therefore, high-intensity training drills produced a sufficient conditioning stimulus with little chance of underrecovery for the performance measures we tested. Countermovement jump and S-RPE may be more sensitive performance recovery indicators.

Key Words Borg CR-10 scale, interval training, power, small-sided games, speed

Introduction

The game of soccer requires players to be effective technically, biomechanically, tactically, psychologically, and physiologically. From a fitness coach standpoint, the areas of psychology and physiology of the athletes are of prime interest. A soccer training session must stress the player's physiological development along with technical and tactical development to prepare the player to cover 9–12 km of movement and to perform approximately 1,000–1,400 short activities such as short accelerations, jumping, and changes of direction with changes in activity occurring every 4–6 seconds during a 90-minute match (5,32). The number of high-intensity sprints is thought to be a separator of a good vs. a great player, with the most effective players performing more high-intensity sprints (4). A high maximal oxygen uptake ($\dot{V}O_2max$) is thought to be an important indicator of a player's ability to perform multiple high-intensity sprints (12). Although some authorities argue that $\dot{V}O_2$ is not an applicable measure of soccer performance, it has been shown to be positively related to work rate. A recent study using male players, showed a significant increase in the number of sprints, involvement with ball, and distance covered with an increase in $\dot{V}O_2max$ (13). Although the majority of research has been conducted using male players at various levels of soccer, Stolen et al. (32) contend that female soccer players should have a lower level of ability, in terms of endurance and strength, when compared to male players.

The methods for improving soccer players' $\dot{V}O_2max$ consist mainly of high-intensity intermittent interval work, either with or without a ball. High-intensity intermittent interval training (defined as repeated bouts of exercise ranging from 10 seconds up to 5 minutes completed at an intensity greater than anaerobic threshold) has been suggested to have a positive impact on $\dot{V}O_2max$ (24). Studies have reported dramatic increases in $\dot{V}O_2max$ using a regimen consisting of 4 sets of 4 minutes of work at an intensity of 90–95% of HRmax with 3 minutes of active recovery at 50–60% of HRmax, with and without the ball (13,15). A founding principle guiding the mode of training is that it incorporates soccer-specific training. Soccer-specific training should mimic the demands of a game and, if possible, involve the ball (15). The use of a ball allows a player to improve tactical and technical skills along with improved fitness.

Although high-intensity interval training has a positive effect on a soccer player's physical fitness and soccer performance, limited data are available regarding the minimal

Address correspondence to Jesper Sjökvist, jesper.sjokvist@sok.se.

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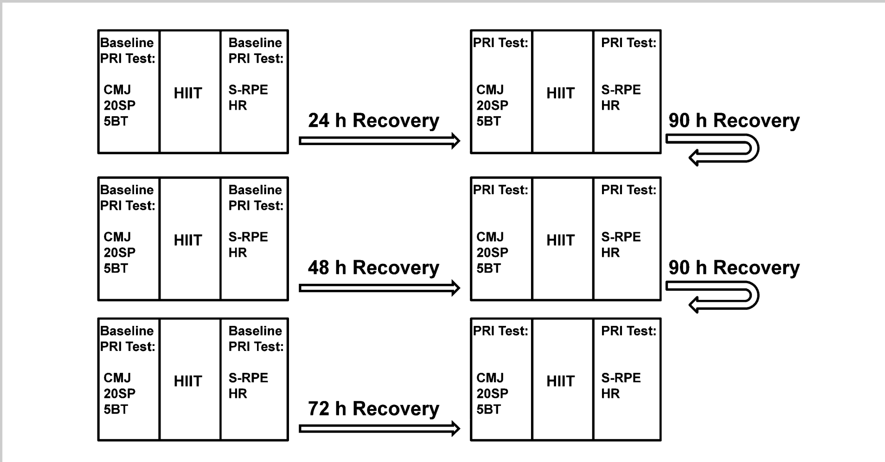

Figure 1. Recovery between sessions (counterbalanced order). PRI = Performance Recovery Indicator, HIIT = high-intensity interval training.

recovery time between exercise sessions, because current research is focused on recovery between repetitions within a training session and not between the training sessions (12). Recently, Bishop et al. (6) stated that the majority of what is known about between-session recovery is largely hypothetical in nature. Further, there is a lack of dependable, noninvasive methods for assessing an individual's level of recovery between sessions. This is problematic as soccer leagues and tournaments call for intense playing schedules with short recovery periods between matches; when to apply high-intensity training sessions without affecting match performance has not been investigated. For example, World Cup matches are typically played every 4–5 days (3), whereas women in NCAA division I routinely schedule 2 matches played over 3 days followed by 1 day of recovery and 3 days training preceding the next match. Understandably, for an NCAA Division I women's soccer coach, finding a formula to plan training efficiently throughout the week between the matches is difficult at best. Indeed, the coach must balance training aimed at improving a player's power, speed, cardiovascular capacity, psychological status, and skills throughout the season, without affecting the player's recovery and readiness to play the next match. Because high-intensity interval training using small-sided games or soccer-specific running tracks has been shown to mimic the motion patterns of a match and stress the cardiovascular system at a similar rate, there is potential for the same experience of fatigue as after a match (11). Analysis of a player's performance in, for example, speed, power, perception of muscle soreness, and oxidative stress after a match has been negatively affected for up to 48 hours post match (1,10,22). Creating a simple and applicable test package that measures recovery of performance indicators would allow for proper alteration of day-to-day training intensities and volumes, maximizing the potential for positive adaptation to training, avoiding overtraining and

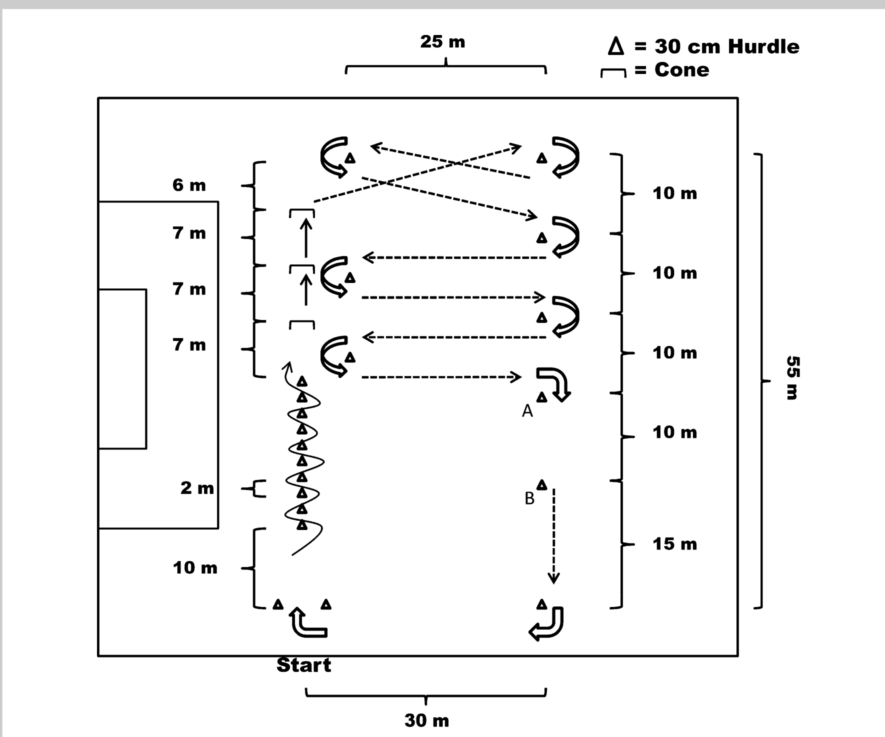

Figure 2. A soccer-specific track for training high-intensity aerobic intervals (after Hoff et al. [15]). Players run following the track indicated by the arrows. Backward running occurred between cones A and B.

TABLE 1. Training drills.

Training drills	Dimension and distance
4 vs. 4	32 × 22.5 m
Soccer-specific running course with and without the ball	~ 282 m per lap

injury, and more importantly, improving match performance. Consequently, there is a need for research that focuses on recovery time and field-based performance recovery indicators such as speed, power, cardiovascular response and psychological status between high-intensity interval training sessions.

To our knowledge, this is the first study to investigate the recovery demands for the physiological and psychological stress encountered from a high-intensity interval training session in soccer. Therefore, the purpose of this study was twofold: (a) to investigate the recovery time required after high-intensity interval training sessions with and without the ball, to recover performance in power, speed, cardiovascular response, and psychological strain and (b) to investigate the sensitivity of the selected performance recovery indicators. Because of pilot work in our laboratory in recovery from running and weightlifting, we hypothesized that performance recovery time would be at least 48 hours after a training bout.

METHODS

Experimental Approach to the Problem

To assess recovery after 24, 48, or 72 hours of rest after high-intensity interval soccer training sessions, we decided to investigate the performance change in speed, power, cardiovascular response, and psychological status. The definition of a recovered subject in our study was an individual who replicated or exceeded baseline performance in the selected performance measures after a high-intensity training session with 24, 48, and 72 hours of rest, regardless of any other perturbations to homeostasis. The selection of these performance measures was based on the subject's familiarity with the tests to optimize the sensitivity of the testing protocol and because they could easily be replicated by coaches and athletes with simple and noninvasive methods. We are aware that recovery can be measured through more invasive methods such as blood sampling and muscle biopsies to assess hormonal, enzymatic, neural and muscle tissue status that may affect the player's homeostasis; however, those methods are not available to the majority of coaches or athletes on an everyday basis.

In total, subjects performed 1 maximal treadmill running test and Yo-Yo intermittent recovery test (Yo-Yo IR1) and 3 series of 2 training sessions (baseline and recovery period) during the study. The training sessions were structured identically and consisted of a standardized warm-up, performance testing, specific and nonspecific soccer interval training, and standardized cool-down. Each series of 2 training sessions consisted of a baseline trial and training session separated by 24, 48, or 72 hours of recovery in a counterbalanced order (Figure 1). Subjects were afforded at least 90 hours of recovery between the series of training sessions.

Each training session began with a standardized dynamic warm-up (~15 minutes) in which light technique work with a ball was performed. Immediately after the warm-up, the subjects performed 2 repetitions of performance tests that consisted of countermovement jumps (CMJs), 5 bounds for distance (5BT), and 20-m sprint (20SP). After performance testing, high-intensity training was performed by using 4 games of 4 vs. 4 small-sided games and 4 sets of interval running without a ball on a soccer-specific running course (Figure 2). Each repetition within the high-intensity exercises was performed for 4 minutes with 3 minutes of active recovery between games or sets. In addition, the small-sided games and the interval running without the ball were separated by up to 5 minutes of active recovery. The dimensions for each high-intensity drill are presented in Table 1. The end of the session was concluded with a standardized cool-down consisting of 5 minutes of light jogging followed by 10-minute stretching. The subjects provided a session rating of perceived exertion (S-RPE), approximately 20–30 minutes after completion of each training session to assess psychological perception of the session. Heart rate (HR) data were collected and analyzed for time spent in medium >80% and high-intensity >90% HR zones (Tzone) and for mean HR during 4 vs. 4 small-sided game and dribbling course drill performed to assess the cardiovascular response to the training session. To ensure reproducibility of the tests, the performance recovery assessment and high-intensity training sessions were held at the same time of the day. Furthermore, the subjects did not perform any organized training and were instructed to refrain from any strenuous activity during the recovery intervals.

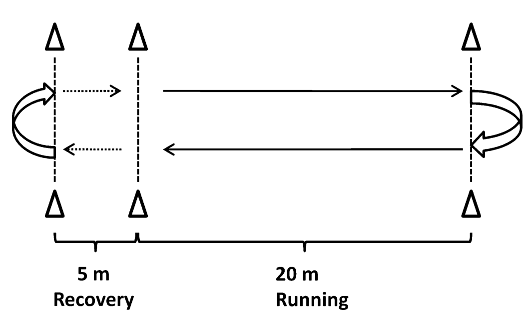

Figure 3. Yo-Yo intermittent recovery test (Bangsbo and coworkers [20]).

Table 2. Participant characteristics ($n = 14$).*

Age	Height (cm)	Body mass (kg)	Body fat (%)	$\dot{V}O_2max$ ($ml \cdot kg^{-1} \cdot min^{-1}$)	HRmax$\dot{V}O_2$max.test ($b \cdot min^{-1}$)	HRmax Yo-Yo IR1 test ($b \cdot min^{-1}$)
20.3 ± 2.3	168.1 ± 4.7	61.9 ± 6.5	20.9 ± 3.4	53.9 ± 5.7	192.1 ± 3.4	194.9 ± 4.1

*Values are given as mean ± *SD.*

Moreover, the subjects were instructed to maintain a normal diet during the recovery interval and reproduce their energy intake and hydration 24 hours before testing.

Subjects

Fourteen ($n = 14$) NCAA Division I female soccer players participated in the study. Each participant's age, height, weight, and %body fat were assessed. Body fat was assessed using skinfold caliper at 3 locations (triceps, suprailiac, and thigh) (17). The subjects had competitive soccer experience ranging from 6 to 10 years and had 1–3 years of experience playing at the Division I collegiate level. The subjects were members of a team competing in the NCAA Southeastern Conference and were ranked in the top 120 of 318 participating schools. Before the investigation, the subjects had participated in 3 weeks of spring off-season strength and conditioning training 4 times per week. The subjects had all undergone extensive medical evaluation as part of the preparticipation regiment required by the University athletics department. All subjects were fully informed of all experimental procedures and risks before giving their written informed consent to participate. The study was approved by the University's Institutional Review Board.

Small-Sided Game and Soccer-Specific Running Track. The subjects were informed of the drill structure and the aim of improving skills and fitness was emphasized to ensure maximum effort. Coaches were present to provide verbal encouragement and to supply balls into the drill to prevent any stoppage during the effective drill time. In the small-sided game, small goals were set up on each team's half and the participants were encouraged to score, but the score was only recorded if the whole team was in the opposition's half of the pitch when the ball crossed the goal line. The soccer-specific running track (Figure 2) was performed without a ball. This course was suggested by Hoff et al. (15) and has been shown to elicit HRs over 90% of HRmax in men. Both the 4 vs. 4 small-sided games and the soccer-specific running track drills were performed for 4 repetitions of 4 minutes with 3 minutes of active recovery between each repetition. To enhance lactate recovery, active recovery consisted of slow jogging or walking with the objective of keeping the HR between 50 and 70% of HRmax (14).

Yo-Yo Intermittent Test 1 and Treadmill $\dot{V}O_2max$ Test. To determine the subjects' work capacity and maximum HR, the Yo-Yo IR1 and treadmill $\dot{V}O_2$max test were performed on separate days. The Yo-Yo IR1 consisted of repeated 2 × 20-m runs between the starting, turning, and finishing lines (Figure 3). The speed progressively increased and was controlled by audio beeps from a CD player. After each running bout, the participants had 10 seconds of active recovery consisting of a 2- × 5-m jog (2). When a subject

Table 3. Mean effect of recovery duration interval on recovery indicators.*†

	S-RPE	Tzone (80–100% HRmax)	20SP (s)	5BT (m)	CMJ (cm)
Baseline	7.9 ± 0.4	1,687 ± 234	3.59 ± 0.17	10.1 ± 0.7	48.8 ± 7.9
24 h	8.4 ± 0.5‡	1,680 ± 141	3.62 ± 0.18	10.2 ± 0.6	46.9 ± 7.6‡
Baseline	8.4 ± 0.5	1,738 ± 135	3.58 ± 0.19	10.3 ± 0.7	48.8 ± 6.6
48 h	8.4 ± 0.6	1,742 ± 137	3.62 ± 0.18	10.3 ± 0.7	48.7 ± 7.9
Baseline	8.6 ± 0.6	1,692 ± 68	3.61 ± 0.19	10.1 ± 0.8	48.2 ± 8.1
72 h	8.3 ± 0.7	1,668 ± 156	3.61 ± 0.21	10.2 ± 0.7	49.3 ± 8.3

*S-RPE = session RPE; 20SP = 20-m sprint; CMJ = Countermovement jump; 5BT = 5 bound jump; Tzone = HR 80–100% in seconds.
†Values are given as mean ± *SD.*
‡Significantly different from baseline at $p < 0.05$ level.

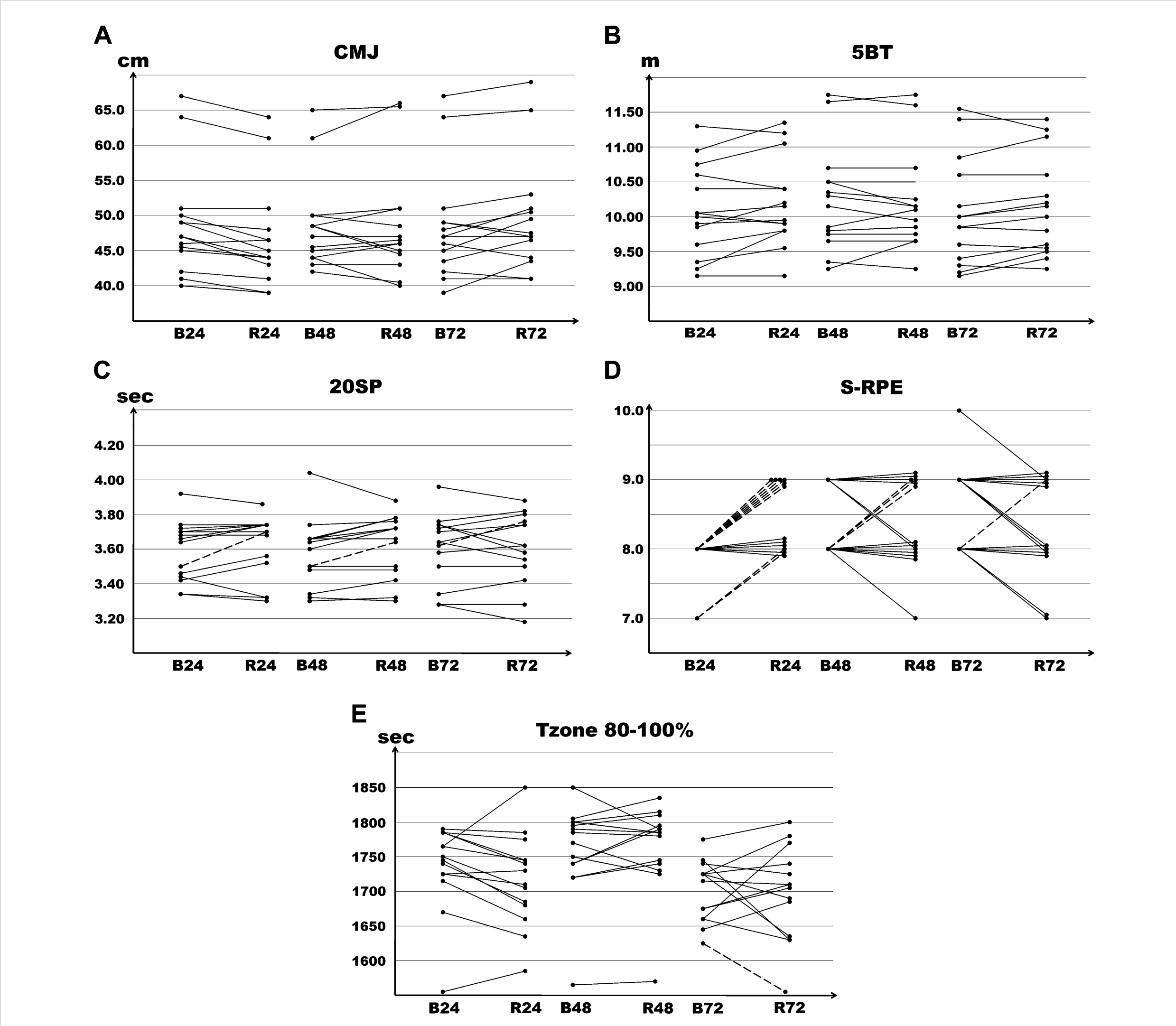

Figure 4. Individual changes in (A) countermovement jump (CMJ), (B) 5 bound jump (5BT), (C) 20-m sprint (20SP), (D) session RPE (S-RPE), and (E) HR 80–100% in seconds (Tzone 80–100%) from baseline and recovery interval 24, 48, and 72 hours. (- - -) Unrecovered according to least-significant difference.

failed to reach the finishing line in the allotted time on 2 consecutive occasions, the test ended and total distance covered and finishing HR were recorded. The finishing HR was used as a measure of maximal HR (HRmax). The HRmax collected during the Yo-Yo IR1 test was used to determine HR zones. All subjects were familiar with the Yo-Yo IR1 test as it was frequently employed in the participants' pre and off-season training regimen.

Treadmill $\dot{V}O_2$max tests were also performed to obtain a $\dot{V}O_2$max using a continuous exhaustive incremental test on a motorized treadmill (Quinton, Seattle, WA, USA). The HR and $\dot{V}O_2$ were measured throughout the protocol. $\dot{V}O_2$ was measured using an automated metabolic measurement system (VacuMed model MX, Ventura, CA, USA). Breath-by-breath gas exchange was recorded every 10-second interval throughout test. The HRmax and $\dot{V}O_2$max were determined as the highest attained value at the time of exhaustion or voluntary termination. Criteria for a maximal test were threefold. The subject should have reached a respiratory exchange ratio of 1.1, an HR of >95% of HRmax as calculated by 220 − age, and report at least 9 on the 10-point Borg CR-10 scale.

The HR during Yo-Yo IR1 and treadmill test was measured at 5-second intervals using a Polar System (Polar OY, Kempele, Finland). Based on HRmax obtained from Yo-Yo IR1 test, each subject's HR zones were calculated corresponding to 80–90 and 90–100% of HRmax.

Performance Recovery Indicator Tests. To determine the level of recovery after 24, 48, and 72-hour rest, 2 power tests, 1 speed

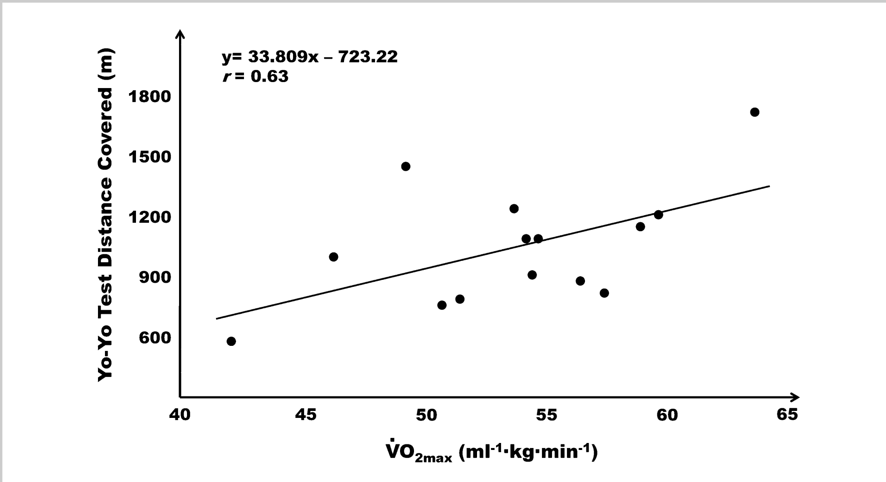

Figure 5. Relationship between Yo-Yo test distance covered and treadmill $\dot{V}O_2$max ($n = 14$).

global difficulty of the entire exercise session. It was emphasized for the individuals to not focus on any particular part of the exercise session (i.e., beginning or ending), rather, they should address the feeling of exertion as a result of the entire session.

test, session HR, and S-RPE were collected. Power and speed were measured immediately after a 15-minute standardized warm-up. The first power test was assessed by performing 2 repetitions of CMJ on a switch mat (Robotics Inc., Huntsville, AL, USA); with the better of the 2 jumps in centimeters recorded. The subjects were instructed to jump as high as possible with the arms actively involved (8). The second power test was determined by a 5BT. Subjects were instructed to stand with their preferred foot forward at the start of the tape measure. The subject then bounded 5 times by alternating left and right feet for the furthest distance possible in 2 trials (9). The furthest distance in meters was recorded. Sprint speed was determined by performing 2 20SPs from a standing start using an electronic timing system (Brower, Salt Lake City, UT, USA). The subjects were instructed to start the sprint on their own volition and run as fast as possible between 2 photo cells. The fastest time in seconds, recorded to nearest hundredth of a second, of the 2 trials was recorded.

Session HR response was measured by determining the difference in time spent in medium-to high-intensity HR zones (i.e., Tzone 80–90 and 90–100% of HRmax) between sessions after recovery durations of 24, 48, and 72 hours. The mean HR for the total training session was also recorded. Rating of perceived exertion (RPE) was collected based on a 0–10 Borg's CR-10 scale (7). After each repetition of each 4 vs. 4 small-sided games and the dribbling course without the ball, subjects provided an RPE to assess the acute rating of perceived exertion. Each individual provided an S-RPE score approximately 20–30 minutes after the last repetition of the high-intensity drill. The purpose of the delay was to prevent potential influence of the last high-intensity bout on the total S-RPE. A thorough explanation of how the scale was interpreted was given before any testing. Each individual was instructed to use the S-RPE scale to subjectively rate the

Statistical Analyses

Descriptive characteristics were reported for age, height, weight, HRmax in Yo-Yo IR1 test, HRmax and $\dot{V}O_2$max on treadmill, and body fat%. In order to detect any significant main effect between recovery and change in performance between session performed at 24, 48, and 72 hours, a series of 2 (trial) × 3 (recovery periods) doubly repeated-measures analysis of variance (ANOVA) for Tzone 80–100% of HRmax, S-RPE, CMJ, 20SP, 5BT, ΔCMJ, Δ20Sp, and Δ5BT was performed. When appropriate, post hoc Bonferroni follow-ups were performed to identify where significant differences occurred.

To determine individual recovered and unrecovered subjects according to the prescribed regimen in this study, inferential statistics were employed to pre-determine criteria based on the least-significant difference for our sample. A least significant difference for CMJ, 20SP, 5BT, S-RPE, and Tzone 80–100% HRmax for 24-, 48-, 72-hours recovery was determined based on a power analysis of 0.8 using a sample size of 14 for a 2-tailed, paired t-test at $p \leq 0.05$ (25). Any subject that failed to perform within the least-significant difference of their initial score was considered unrecovered.

Additionally, the relationship between distance covered in the Yo-Yo IR1 and the $\dot{V}O_2$max treadmill test was examined through a Pearson product–moment correlation ($n = 13$). All data were analyzed using SPSS 16.0 Software (SPSS Inc., Chicago, IL, USA). Intraclass coefficients were used to test reliability of the tests used in the investigation. Statistical power was determined to range from 0.70 to 0.85 for the sample size used with the variables measured in this study. Statistical significance was determined a priori at the 0.05 level. All data are reported as mean ± *SD*.

Results

The subjects' general descriptive data are presented in Table 2. The mean total time to complete each testing session was 88 ± 2.5 minutes, which included (a) warm-up, (b) performance recovery indicator tests, (c) 4 vs. 4 small-sided games of 4 minutes with 3 minutes of rest between repetitions (d) 4 × 4 minutes with 3-minute rest between repetitions of running on a soccer-specific track, and (e) a cool-down. The final small-sided game and the soccer-specific running track drill were separated by an average of 4.0 ± 0.2 minutes of active recovery (50–70% of HRmax). In addition, the test–retest reliability of the performance recovery indicators yielded an ICCM coefficient ranging from 0.93 to 0.99. These values seem to indicate a high level of reliability of the specific markers employed in measuring performance.

Performance Recovery Indicators

Power and Speed. Results from the repeated-measures ANOVA revealed no significant differences ($p > 0.05$) in 20SP or 5BT performance between each baseline and the trials after 24, 48, or 72 hours of recovery. After 24 hours of recovery, the CMJ was lower ($p < 0.04$) than baseline, but no difference was detected in CMJ between baselines and 48 and 72 hours of recovery (Table 3). When comparing the change in performance between the training sessions after 24, 48, or 72 hours of recovery, no significant differences were detected in Δ20SP or Δ5BT between training sessions. There was a significant difference for the ΔCMJ after 24-hour recovery compared to the 48- and 72-hour recovery interval. Subjects jumped lower in the training session after 24-hour recovery compared to 48 hours of recovery ($p < 0.05$), and higher in the training session after 72 hours of recovery compared to the 24-hours recovery interval ($p < 0.01$). There was no difference in ΔCMJ between the 48- and 72-hours recovery sessions.

Heart Rate Performance. The subjects' mean session HR for was 76.7 ± 4.3% of HRmax. Subjects spent on average 23.4 ± 1.1 minutes at +90% of HRmax during training sessions. There was no significant difference detected in mean session HR between baseline and each recovery trial at 24, 48, or 72 hours. Also, no differences were found regarding mean HR during 4 × 4 minutes of 4 vs. 4 small-sided games or 4 × 4 minutes of the soccer-specific running track drill during the baseline sessions and either 24, 48, or 72 hours of recovery. There were no differences in Tzone 80–100% of HRmax when combining 4 vs. 4 small-sided games and soccer-specific running track drill between baseline and either 24, 48, and 72 hours of recovery (Table 3). When Tzone 80–100% was compared for training sessions performed after 24,48, and 72 hours of recovery, subjects spent more time in the Tzone 80–100% in the training sessions after 48 hours of recovery compared to the training session after 24 hours of recovery ($p < 0.01$) and 72 hours of recovery ($p < 0.01$), respectively.

Rating of Perceived Exertion. The S-RPE values were not significantly different between baseline and 48 and 72 hours of recovery. However, S-RPE values were higher for the training session after only 24 hours of recovery ($p = 0.02$) compared to baseline (Table 3). There was no difference in S-RPE among the training sessions performed after 24, 48, or 72 hours of recovery.

Individual Data Analysis. Individual data are presented in Figures 4A–E. The least-significant difference accepted for CMJ, 20SP, 5BT, S-RPE, and Tzone 80–100% HR zone was 5.8 cm, 0.13 seconds, 0.6 m, 0.6, and 119 seconds, respectively. When analyzing CMJ data, despite there being a significant reduction in CMJ for group means, the individual scores were all just within the least-significant limits, resulting in all subjects appearing fully recovered after 24, 48, and 72 hours. In 20SP, only 1 subject did not recover after 24 and 48 hours, and another subject did not recover after 72 hours. However, all subjects' performances were recovered in the 5BT after all recovery durations. Tzone 80–100% HR zone yielded 1 unrecovered subject after 72-hour recovery, though all subjects recovered after 24 and 48 hours of recovery. Analysis of S-RPE revealed 8 unrecovered subjects after 24 hours, 3 unrecovered after 48 hours, and 1 after 72 hours of recovery. Two subjects who were unrecovered after 24 hours were also unrecovered after 48 hours of recovery. One subject who was unrecovered after 24 hours was also unrecovered after 72 hours of recovery.

$\dot{V}O_2max$ Treadmill Test vs. Yo-Yo IR1 Test. The $\dot{V}O_2$max treadmill test elicited HR values approaching 98.6 ± 0.8% of HRmax registered during the YoYo IR1 test. There was a moderate correlation ($r = 0.63$) between distance covered in the Yo-Yo IR1 test and the $\dot{V}O_2$max measured on the treadmill (Figure 5).

Discussion

The aims of this study were twofold: (a) to investigate the performance recovery interval needed after high-intensity interval training sessions with and without the ball to fully recover power, speed, cardiovascular response, and psychological strain in amateur women soccer players and (b) to investigate the sensitivity of selected field-based performance recovery indicators. Based on pilot work performed in our laboratory in recovery from running and weightlifting, it was hypothesized that the performance recovery time interval would be at least 48 hours. Overall, this study demonstrated that women soccer players appeared fully recovered, except for CMJ (power) and S-RPE, in 20SP, 5BT and session HR response within 24 hours of high-intensity interval training with and without the ball. After 48 hours, CMJ and S-RPE were also recovered.

It should be noted that we purposely limited our definition of training recovery to the ability to replicate or exceed initial performance on our measures. Clearly, we do not say that our subjects were fully recovered, because we recognize that other disruptions to homeostasis such as changes to muscle tissue, neural, and hormonal status may have occurred; however, we choose not to investigate this. Instead, we investigated what we considered to be a practical, performance-based assessment of performance recovery that could be used by coaches and athletes. Likewise, our performance measures were somewhat general and fairly simple, so that performance recovery could be readily assessed by coaches and also perhaps be applicable to other sports.

The only recovery indicators significantly affected from baseline levels after 24 hours of recovery were CMJ and S-RPE. The subjects jumped significantly lower, and the perception of exertion was increased after 24 hours of recovery. A similar response in vertical jump height was reported by Andersson et al. (1) who showed a decrease in vertical jump heights up to 48 hours after a competitive soccer match. Because this study's training session did not equal the same stimulus as a full 90-minute competitive game but still affected the CMJ negatively after 24 hours, the training stimulus from 4 × 4 minutes of high-intensity small-sided games and soccer-specific running track drill appear to potentially reduce aspects of neuromuscular capacity for 24 hours. However, a recent study found the opposite response regarding sprint and jump performance after a match in the Women's Danish Premier League, where sprint ability was affected while jump performance was not compromised (22). Thus, it seems that jump performance after matches and high-intensity training shows contrasting results. Therefore, it is important for the coach to collect data over time to produce a profile for each player's neuromuscular response to training sessions or matches. Interestingly, along with a decrease in vertical jump height, one would have expected a decrease in 20SP and 5BT, because all 3 recovery indicators depend on strength and explosive power. However, we did not observe this.

Wisloff et al. (33) reported a strong correlation between strength and performance in both sprints and vertical jumps, which would support the presence of fatigue in both exercises. There might be a difference in activation patterns between the 2 exercises, which might affect the amount of muscle work and intramuscular coordination and thus the time needed for recovery (16,18). Furthermore, a study in our laboratory (23), in which female subjects performed 3 sets of 8 maximal 30-m sprints with 45 seconds between repetitions and 5-minute recovery between sets, revealed no difference in sprint performance after 24, 48, or 72 hours of recovery. This study further supports this study because short maximal sprints are integral to soccer performance. To the best of our knowledge, no study has investigated the activation pattern for 5-bound jumps. It seems plausible that 5BT has similarities with sprinting in its performance, which might support the lack of effect in this recovery indicator in this study. Coutts et al. (9) found that the 5-bound jumps test was the most sensitive to high-intensity exercise. Our data would suggest that CMJ and S-RPE might be more sensitive predictors of underrecovery in our study sample and conditions.

Session RPE was significantly higher after 24 hours of recovery compared to baseline. However, it did not seem to have a negative effect on the majority of performance recovery indicators, because only CMJ was affected throughout the recovery trials. The subjects managed to achieve the same cardiovascular response to the 4 × 4-minute small-sided games and soccer-specific running track drill and performance in 20SP and 5BT remained unaffected. Comparing to other intermittent sports, this response is in agreement with Kraemer et al. (19), who found a limited effect on physical recovery but reported perceptual fatigue after a 24-hour recovery period from a tennis match in female players. The S-RPE might be a sign of early accumulation of fatigue as S-RPE after 24 hours was determined after completion of the second identical training session. That is, if another identical session were performed again within 24 hours, additional recovery indicators might have been affected. This is supported by several studies stating that changes in RPE occur before any changes in physical recovery indicators, which may make RPE a more sensitive indicator for adaptation or maladaptation to training (27,28,31). Andersson et al. (1) showed peak muscle soreness after 27 hours and took up to 67 hours to return to baseline values. Muscle soreness can occur in high-intensity training such as soccer and has been linked to structural alterations within the muscle (29). As muscle soreness increases, it could impact RPE because RPE takes into account the physiological effort and the psychological state of the athlete. If an athlete perceives her body to be painful and stiff during high-intensity training, it would seem plausible that RPE may increase. However, in this study, no assessment was made regarding the subjects perceived soreness after the recovery intervals.

Individual data were analyzed in this study with regards to various indicators of recovery. Based on a calculated least-significant difference, all subjects were recovered based on the 5BT, CMJ, and Tzone 80–100% of HRmax after 24, 48, and 72 hours of recovery. Regarding the 20SP, 1 subject after 24 hours, 2 subjects after 48 hours, and 1 subject after 72 hours of recovery were not fully recovered. S-RPE indicated that 8 subjects were unrecovered after 24 hours, 3 subjects after 48 hours, and 1 subject after 72 hours of recovery. These analyses allow the coach an opportunity to adjust training volumes and intensities to each individual player to avoid overreaching or in the worst cases, overtraining.

The limited impact on recovery from these high-intensity training sessions may suggest that the sessions were not sufficiently physically demanding. However, the subjects spent an average of 23.4 minutes at +90% of HRmax during these sessions, and their S-RPEs were 8 or higher. From previous experience with male soccer players in an English Championship league club, a training session lasting up to 90 minutes, in which 15 minutes were spent at +90% of HRmax was considered a highly physically demanding session in soccer. Furthermore, S-RPE in the mid 8s indicated that the players perceived the training sessions at being close to maximal. Therefore, we considered these sessions to be very taxing.

The traditional method of measuring work rate, HRmax, and oxygen uptake has been in laboratory settings using either bike ergometers or motorized treadmills (26). Although these tests are useful in assessing a player's physiological capacity, they are not sport specific, time efficient or cost effective, which has led to the development of soccer-specific field tests. The most popular test used over the last 1015 years is the Yo-Yo IR1 test. Developed by Bangsbo in the mid-1990s, it has been correlated with $\dot{V}O_2max$ ($r = 0.71$) as tested on a motorized treadmill, the number of high-intensity sprints performed in a soccer match ($r = 0.71$), higher work rate in women players ($r = 0.56$), and passing accuracy ($r = -0.65$) (20,21,30). Research focusing on the validity of the Yo-Yo IR1 test has been performed mainly on elite level male soccer players and may not exhibit the same relationship in collegiate amateur female players. Consequently, there is a need for research focusing on the validity of the Yo-Yo IR1 test in amateur women soccer players compared to laboratory treadmill testing of work capacity.

In this study, the treadmill $\dot{V}O_2max$ test and the Yo-Yo IR1 test were only moderately correlated ($r = 0.63$) when comparing $\dot{V}O_2max$ (ml·kg^{-1}·min^{-1}) and distance covered (in meters) in the respective tests. These findings confirm previous results who reported a correlation of $r = 0.71$. It is important to point out that, although our findings are in agreement with other studies, our statistical power was 0.7 that is possibly because of limited utility to correlations yielded with samples sizes <25. Furthermore, peak HRs obtained during the $\dot{V}_2max$ treadmill tests were 98.6 $\pm$ 0.8% of HRpeak observed during the Yo-Yo IR1 test. These results are the reverse of those reported previously who found that HRpeak obtained during the Yo-Yo IR1 test were 99 $\pm$ 1.0% of those obtained during the $\dot{V}_2max$ treadmill test in men (20). That is, lower HRpeak values were reported during the Yo-Yo IR1 test, whereas this study found lower HRpeak during treadmill testing. However, these differences are negligible as they represent $\pm$ 1% in either direction between protocols. Ultimately, these findings suggest that women soccer players respond similarly to men when using the $\dot{V}_2max$ treadmill test and the Yo-Yo IR1 test to evaluate peak HRs. However, it should be pointed out that although Krustrup et al. (20) found that the $\dot{V}_2max$ and distance covered in the Yo-Yo IR1 test were moderately correlated, only the Yo-Yo IR1 test was positively correlated to high-intensity efforts during a match.

Practical Applications

In summary, this study demonstrated that high-intensity interval training with and without the ball using the 4 $\times$ 4 minutes with 3 minutes of active recovery protocol had a limited effect on measured speed, power, and HR response in subsequent training sessions with 24 hours of rest, as measured by our tests, in women soccer players. However, power (i.e., CMJ) and psychological state (S-RPE) should be considered when using this type of training regiment on consecutive training days as these may take longer to recover or may be more sensitive to underrecovery. In our sample of players, employing this type of training appears plausible close to a match with limited negative effects on the player, but recovery should be assessed at different time points in the season and after consecutive sessions with similar stimuli, and the coach should document and create a profile for each player's response to training and matches. We believe these performance recovery indicators are simple and applicable for a team with limited resources. However, the indicators do not assess a player's recovery with regard to muscular tissue, hormonal or neural status, but rather a player's ability to reproduce or exceed performance in power, speed, cardiovascular response, and psychological perception after high-intensity interval training, because these qualities are important for soccer. In addition, distance in the Yo-Yo intermittent recovery test IR1 may be used as an estimate of maximal work capacity for women soccer players because it is moderately correlated with $\dot{V}_2max$ treadmill test. Future research should focus on potential gender responses to sport-specific and nonspecific high-intensity exercises, because men might recover differently. Moreover, research should also focus on the effect of recovery on multiple high-intensity sessions.

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